Board to Be Wild
Engineering Design Report

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1 Introduction

Engineers will create personalized cutting board designs using fusion 360 or adobe, then students will create tool paths for their boards using aspire.My purpose of this project was to create a Wanted poster of a pirate from a TV show called one piece which is my favorite show at the moment. I plan to take my board and hang it in my new room because we are moving.

2 Initial Design and Planning

My design is based on the anime One piece and the main character Monkey. D . Luffy, who is on a mission to become the king of the pirates. As he faces more foes and battles against the higher-ups in the marines his bounty begins to rise, and the most notorious pirates always have the highest of bounties.

Figure 1: Inspiration for my design

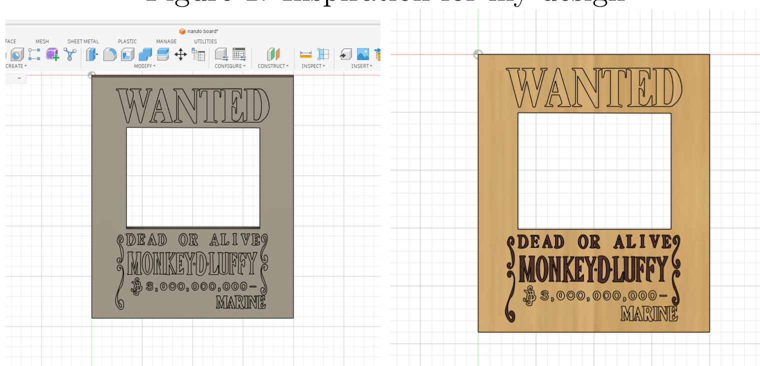

Figure 2: Creation of the design in fusion 360

3 Material Preparation

when starting our project, we started with two pieces of wood one cheery and one being maple we then worked with Mr.Budzichowski to cut our pieces of wood into 2x10 inch blocks that we would then glue together. During the gluing process, students needed to be very precise with how much glue was poured and making sure that no pockets were left in between the blocks that could potentially mess up the milling process. Thomas, who was already done gluing, helped me pour, although we made a complete mess, we were able to successfully glue my blocks without creating any pockets in my blocks.

Figure 3: Me and Thomas working with the glue

Figure 4: after sticking all pieces together to dry

4 Digital Design

The digital design for my charcuterie board followed a simple but tedious procedure first with the text wanted and the name of the fugitive monkey D. Luffy First i used a screenshot of the image to properly scale and replicate the lettering using the draw tool in fusion360 it was like tracing an image on a book .but on keyboard and mouse. Then i cut out a square where i would place my mini figure using the cut tool, coming out with perfect measurements that fit my board and my figure.

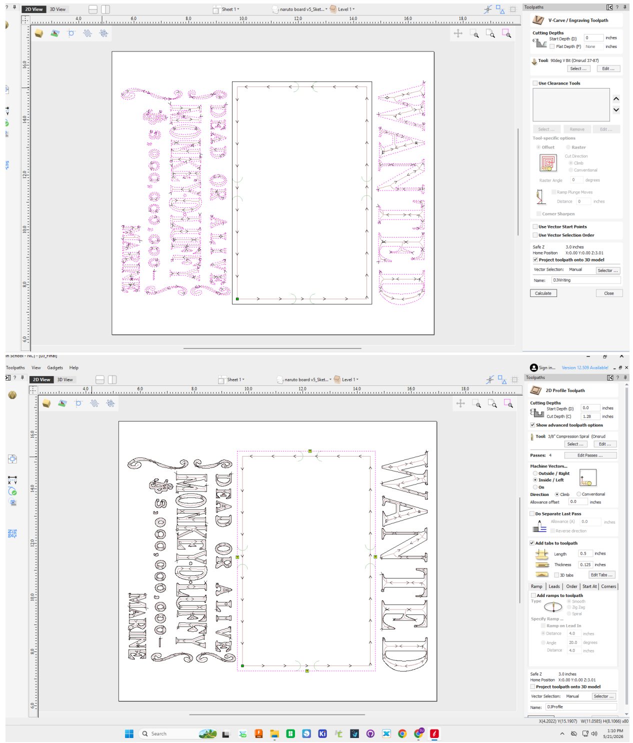

Figure 5: These are the toolpaths for the board

5 CNC and manufacturing

When using the CNC machine we were first tasked with creating the tool path's for our boards so the first process of my tool path was using the 3/8th compression spiral to being my profile cut of the hole I was creating in the board, then the 90 deg tool was used to cut out the The text "wanted Dead or alive Monkey. D Luffy" This process took around 30 minutes due to the machine having to make precise cuts with very tight angles. Videos for the CNC process are Linked in my google drive

Figure 6: The final look of my board after milling

Figure 7: Smoothing out the inside of the board

6 Resin Process

For my charcuterie board the resin process took days since I was completely filling a hole in my board with resin, First i had to tape a cardboard cut slightly larger than my hole to prevent leakage of the resin, then i filled the whole with resin without the mini figure in it pouring in about half an inch of resin into my board, The next step was placing my mini figure inside of the board and pouring a little resin on top to make sure he did not float, Finally the last two days of pouring consisted of me just over flowing my board with resin so it could seep into the letters of the wanted sign. After finishing pouring my resin i had to plane my board, I did this 5 times to get to a good spot where the resin was even with the rest of my board, then I had to begin the sanding process where I had to work my way up from a 60 grit sander all the way to a 220 grit typically using each sander for around 10 minutes.

Figure 8: Here is my 3 day resin process filling around luffy

Figure 9: Final cutting board

7 Final Product

8 Reflection

During this process using resin, CNC machines, and sanding tools were challenging yet also fun Because it required precision,patience, and attention to various details, I think the most challenging part of the project was figuring out how to uses aspire and creating tool paths for the CNC machine. This process was a lot of trial and error causing me to find varous solutions to creating my tool paths. Working with resin was also a new learning point for me since i've never done it before it taught me the importance's of timing and accuracy because incorrect mixing ratios could cause bubbles, uneven curing, or even just imperfections that could change the final look of the board. Sanding the board from 60 to 220 grit was also a new learning experience each level of grit plays an important role in creating a smooth surface. The rougher grits removed imperfections such as uneven wood levels and just enhancing the final appearance of the wood and resin. Throughout this project I improved on understanding how precision and consistency directly affect the quality of the final product.